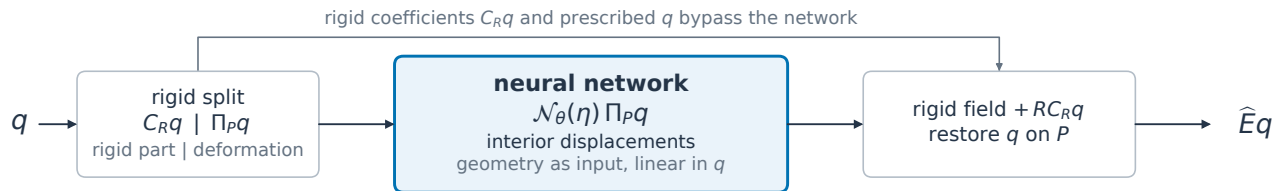


(a) Condensed stiffness of one cell



by construction
(Eq. (9))

$$J_P \hat{E} = I_P$$

q reproduced on P

$$\hat{E} R_P = R$$

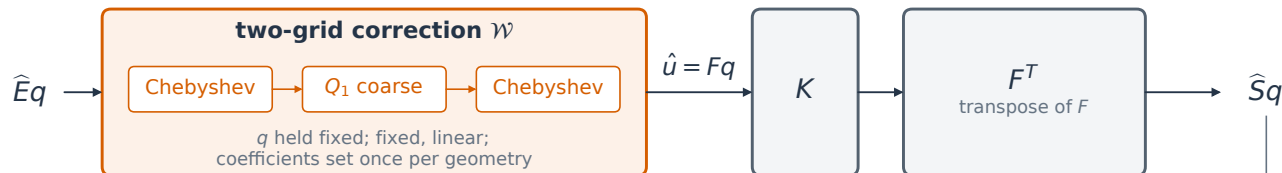
rigid-body motion reproduced

$$\hat{E} \text{ linear in } q$$

at fixed geometry

$$\text{one } \theta$$

shared by all cells



condensed
stiffness

$$\hat{S} = F^T K F$$

symmetric, ≥ 0 , rigid-body null space

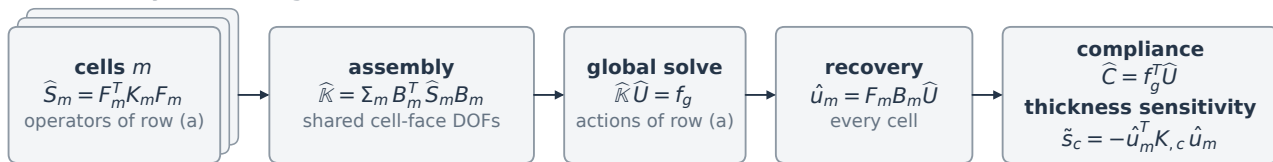
$$\hat{S} - S = H^T A H \geq 0$$

error quadratic in H

$$S \leq \hat{S} \leq \hat{S}_{\text{net}}$$

exact or shifted coarse solve, spectrum in $(0, b]$

(b) Assembly and design



design iteration: update of the corner parameters τ_c (new geometry η , same network parameters θ ; correction coefficients recomputed)